

DATABASE LAB

Lab3: DML

Ruba Sultan



Data Manipulation Language(DML)¹

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- A DML statement is executed when you¹:
 - ▣ Add new rows to a table
 - ▣ Modify existing rows in a table
 - ▣ Remove existing rows from a table

- A Transaction consists of a collection of DML statements that form a logical unit of work¹.

¹ Introduction to Oracle: SQL and PL/SQL P9-3 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

INSERT Statement²

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- Add new rows to a table using the INSERT statement

```
INSERT INTO table [(column [, column,...] ) ]  
VALUES (value [, value...]);
```

- Only one row is inserted at a time with the previous syntax.

² Introduction to Oracle: SQL and PL/SQL P9-5 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

Inserting New Rows³

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- ❑ Insert a new row containing values for each column.
- ❑ List values in the default order of the columns in the table.
- ❑ Optionally list the columns in the INSERT clause.
- ❑ Enclose character and date values within single quotation marks.

Inserting Rows with Null Values⁴

- Implicit method: Omit the column from the column list.

```
INSERT INTO departments (department_id,  
department_name, Location)  
VALUES(280,'Client Relation',2500);
```

- Explicit method: Specify the NULL keyword

```
INSERT INTO departments  
VALUES(290, 'Maintenance ', NULL, 1700);
```

Questions

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? Question 1

Write an SQL statement to add the following information of employee.

ID: **207**

First Name: **Rami**

Last Name: **Ali**

Email: **RamiAli**

Hire Date: **Current Date**

Job ID: **IT_PROG**

Salary: **\$2500**

Ex:

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```
INSERT INTO employees
VALUES      (114,
             'Den', 'Raphealy',
             'DRAPHEAL', '515.127.4561',
             TO_DATE('FEB 3, 1999', 'MON DD, YYYY'),
             'AC_ACCOUNT', 11000, NULL, 100, 30);
```

1 row created.

Copying Rows from Another Table

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- Write your `INSERT` statement with a subquery.

```
INSERT INTO sales_reps(id, name, salary, commission_pct)
  SELECT employee_id, last_name, salary, commission_pct
 FROM   employees
WHERE  job_id LIKE '%REP%';
```

4 rows created.

- Do not use the `VALUES` clause.
- Match the number of columns in the `INSERT` clause to those in the subquery.

UPDATE Statement⁵

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- Modify existing rows with the UPDATE statement

UPDATE tablename

SET column=value[, column=value,...]

[WHERE condition(s)];

- Update more than one row at a time, if required.

Note:

All rows are modified in a table if WHERE clause is not used.

Questions

? Question2

Write an SQL statement to modify salary of employee **106** to be **\$12000**.

? Question3

Write an SQL statement to modify salary and commission of employee **105** to be same as that of employee **155**.

Updating Rows Based on Another Table

- Use subqueries in UPDATE statements to update
- rows in a table based on values from another table.

```
UPDATE copy_emp
SET    department_id = (SELECT department_id
                        FROM employees
                        WHERE employee_id = 100)
WHERE  job_id        = (SELECT job_id
                        FROM employees
                        WHERE employee_id = 200);

1 row updated.
```

DELETE Statement

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- Remove exiting rows from a table using DELETE statement.

```
DELETE [FROM] tablename  
[WHERE condition(s)];
```

Note:

All rows are deleted in a table if WHERE clause is not used.

Deleting Rows Based on Another Table

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- Use subqueries in DELETE statements to remove
- rows from a table based on values from another table

```
DELETE FROM employees
WHERE department_id =
    (SELECT department_id
     FROM departments
     WHERE department_name LIKE '%Public%');

1 row deleted.
```

Database Transitions⁶

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- Consist of one of the following statements:
 - DML statements that make up one consistent change to the data.
 - One DDL statement.
 - One DCL statement.

⁶ Introduction to Oracle: SQL and PL/SQL P 9-24 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

Database Transactions(cont)⁷

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- Begin when the first executable SQL statement is executed
- End with one of the following events:
 - ▣ COMMIT or ROLLBACK is issued
 - ▣ DDL or DCL statement executes(automatic commit)
 - ▣ User exits
 - ▣ System crashes

⁷ Introduction to Oracle: SQL and PL/SQL P 9-25 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

Advantage of COMMIT and ROLLBACK⁸

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- ❑ Ensure data consistency
- ❑ Preview data changes before making changes permanent
- ❑ Group logically related operations

⁸ Introduction to Oracle: SQL and PL/SQL P 9-26 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

Transaction Statements⁹

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□ COMMIT

- ▣ Ends the current transaction by making all pending data changes permanent

□ SAVEPOINT name

- ▣ Marks a savepoint within the current transaction

□ ROLLBACK [TO SAVEPOINT name]

- ▣ ROLLBACK ends the current transaction by discarding all pending data changes.
- ▣ ROLLBACK TO SAVEPOINT rolls back the current transaction to the specific savepoint.

Implicit Transaction Processing¹⁰

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- An automatic commit occurs under the following circumstances:
 - ▣ DDL statement is issued
 - ▣ DCL statement is issued.
 - ▣ Normal exit from application without explicitly issuing COMMIT or ROLLBACK.
- An automatic rollback occurs under an abnormal termination of application or system failure.

¹⁰ Introduction to Oracle: SQL and PL/SQL P 9-28 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

State of the Data Before COMMIT or ROLLBACK¹¹

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- ❑ The previous state of the data can be recovered.
- ❑ The current user can review the results of the DML operations by using the SELECT statement.
- ❑ Other users cannot view the results of the DML statements by the current user.
- ❑ The affected rows are locked, other users can't change the data within the affected rows.

¹¹ Introduction to Oracle: SQL and PL/SQL P 9-29 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

State of the Data After COMMIT¹²

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- ❑ Data changes are made permanent in the database.
- ❑ The previous state of the data is permanently lost.
- ❑ All users can view the results.
- ❑ Locks on the affected rows are released, those rows are available for other users to manipulate
- ❑ All savepoints are erased.

¹² Introduction to Oracle: SQL and PL/SQL P 9-30 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

State of the Data After ROLLBACK¹³

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- ❑ Discard all pending changes by using the ROLLBACK statement.
 - ▣ Data changes are undone.
 - ▣ Previous state of the data is restored.
 - ▣ Locks on the affected rows are released.

¹³ Introduction to Oracle: SQL and PL/SQL P 9-32 Neena Kochhar, Ellen Gravina and Priya Nathan, July 1999

Questions

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? Question4

Write an SQL statement to add the following information of employee.

ID: **208**

First Name: **Rami**

Last Name: **Sameer**

Email: **RamiSameer**

Hire Date: **Current Date**

Job ID: **SH_CLERK**

Salary: **\$1800**

Questions

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Make a permanent change on pervious action

Delete job history for employee **122**.

Modify the location id for department **30** to be **1700**.

Add the following department which has the following information

ID: **300**

Name: Auditing

Location ID: **2500**

Questions

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Rollback the pervious changes and notice the changes before and after rollback.